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The making of an AI news anchor—and its implications

Matyas Bohacek[a] and Hany Farid[b,1]



A CNN anchor was all too easy to deepfake with a few simple AI tools. Image credit: Mike Coppola/Getty Images for CNN.

This summer saw a months-long strike that pitted writers and performers against major Hollywood studios. A particularly fraught point of contention centered around the use (or not) of generative AI (also known as synthetic media or deepfakes). Aiming to find common ground, so that AI and writers can co-exist, the writers' settlement requires that AI-generated material cannot be used to undermine a writer's credit, and its use must be disclosed to writers. Protections for performers include that studios give fair compensation to performers for the use of digital replicas and for the labor unions and studios to meet twice a year to assess developments and implications of generative AI.

As the strike winds down, and as governments consider appropriate worker protections, many continue to grapple with the potential impact of generative AI on creative professions: In particular, what is it, what is it capable of, and what *will* it be capable of?

While entertainers consider the implications and policymakers slowly ruminate over the details, it's already clear that the technology is exceptionally convincing. We—a computer science college student and a computer and information sciences professor—demonstrate here, in collaboration with a major TV network, that it's relatively easy (and getting easier) to create an AI replica of not only a performer, but a figure that people trust to report the truth about world events. One clear takeaway: It's time to stop mulling over future implications and to start considering near-term impacts, as well as potential regulations and controls.

Author affiliations: [a]Electrical Engineering and Computer Science, Stanford University, Stanford, CA 94305; and [b]University of California, Berkeley, CA 94720

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[1]To whom correspondence may be addressed. Email: hfarid@berkeley.edu.

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A Different Kind of Disruptor

Generative AI refers to text, image, audio, or video that has been automatically synthesized by an AI-powered system (1). From only a text prompt, this technology can, for example, write a TV script; from only a text prompt, it can create a photo-realistic image; from only a few minutes of audio, it can clone an actor's voice; and generative AI can insert a person's likeness into anything ranging from a blockbuster movie to a newscast. Even in these early days, this technology can create stunningly realistic and creative content. The next generation will surely improve in terms of power, realism, and accessibility.

Technological disruptions are, of course, not new. What makes this latest technological revolution different than others is that generative AI is trained on massive amounts of human-generated content in order to emulate human creativity (2). Because these systems wouldn't exist without harvesting vast amounts of human-created data, this potential disruption to labor markets is qualitatively different than, for example, the disruption to bank tellers brought about by online banking and ATMs or the disruption to travel agencies brought about by online search and booking.

Mimicking a News Anchor

In collaboration with CNN, we attempted to create the opening of an AI-themed episode of *The Whole Story*, hosted by Anderson Cooper. The short script was not AI generated, but the rest of Cooper was. Here, we describe how, over a two-month period, using primarily open-source software, a first-year undergraduate student (the first author) was able to create an AI news anchor compelling enough to air on network television.

The creation of an AI anchor consisted of two main parts: 1) generate an audio recording of the anchor reading a script; and 2) starting with a previously recorded video clip of the anchor, replace the audio and modify the anchor's mouth and lower face to be consistent with this new audio (a so-called lip-sync deepfake). In describing these two steps, we intentionally omit some of the details as to how we accomplished this, so as not to further equip malicious actors. But we should be clear: None of these details required any significant innovations.

The audio was created by first extracting examples of the anchor's voice from 17 minutes of YouTube videos. These audio clips were used to clone the anchor's voice using ElevenLabs' commercial voice-cloning tool (3). Once cloned, we were able to generate an audio clip in the anchor's voice from a text file containing the desired script.

The deepfake came to life in four sequential steps: 1) The anchor's head was automatically localized throughout each frame of the source video; and 2) a cropped video of just the anchor's head was then provided, along with the generated audio, to the open-source VideoRetalking (4, 5). For each video frame, this neural-based system generated a new mouth region consistent with the new audio; 3) the resulting video was then provided to the open-source CodeFormer (6, 7). For each video frame, this generative adversarial network (GAN) enhanced the quality of the generated mouth by, for example, deblurring and removing obvious misshapen or absent teeth; and 4) the original face in each frame of the source video was replaced with the newly generated face. Lastly, a few preprocessing and postprocessing customizations refined the appearance of the final deepfake. Using this approach, we created an AI-generated version of Anderson Cooper reading parts of the opening of the December 3rd 2023 episode of *The Whole Story*. (See https://cnnpressroom.blogs.cnn.com/2023/11/27/cnns-nick-watt-investigates-ai-and-the-future-of-humanity-for-the-whole-story-with-anderson-cooper/.)



Fig. 1. This news anchor isn't real. AI creations like this one are exceedingly easy to craft. Image credit: "A photo of a trusted middle-aged female news anchor sitting at a news desk," image generated by Stable Diffusion on August 27, 2023 (https://huggingface.co/stabilityai/stable-diffusion-xl-base-1.0).

We did run into some limitations. Large camera or head motions, for example, made it difficult to maintain temporal coherence, and unusual words in the script occasionally led to visually obvious errors in the shape of the mouth. In addition, the type of lip-sync deepfake we created works best when there is a large corpus of videos from which to select. As a result, right now, those with a large digital footprint—like Anderson Cooper—are more vulnerable to having their likeness copied. We have little doubt, however, that these limitations will soon be overcome with the next generation of software tools—tools that, even over the two months we worked on this project, evolved at a surprisingly fast pace.

From Mimicking to Creating

Generative AI can not only copy; it can also produce entirely new content and people. For example, image-based generative AI can produce highly realistic and customizable images of people who never existed. We created the person in Fig. 1 by prompting an open-source image generator (8) with "a photo of a trusted middle-aged female news anchor sitting at a news desk."

Bringing this person—whom we named Sarah Connor—to life required only a few more steps. We next asked ChatGPT to write an opening line for our news anchor, yielding "Good evening ladies and gentlemen. I'm Sarah Connor, and this is your daily dose of Future Flash." We then generated a voice using ElevenLabs customized voice generation (3) with the prompt "middle-aged, women, American, trusted news-anchor," yielding a voice for our news anchor reading the AI-generated script. Lastly, we used the open-source Spline Motion Model (9) to animate the body and head of the single image and then used the same lip-sync deepfake pipeline as described above to bring this news anchor to life. (For audio and video generated, see https://github.com/matyasbohacek/AI-news-anchor.)

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This whole-cloth creation of an AI anchor took us two days to complete. Although the final video is not of the same quality as the AI version of Anderson Cooper, there's little doubt that it will soon be possible to create highly realistic AI performers.

The creation of an AI-generated news anchor is not hypothetical. As early as 2018, China's state news agency Xinhua has been experimenting with AI-powered news anchors (10). There are clear incentives for a network to move from human- to AI-powered anchors. Some seem reasonable, such as the appearance and voice of AI anchors being customized to appeal to different markets. Other implications are more worrisome, as networks may simply decide to prioritize profits and eliminate their human workforce or push consistent and customized state-sanctioned narratives.

From Lost Jobs to Disinformation

Generative AI is also being weaponized by disinformation peddlers who co-opt the likenesses of trusted anchors and celebrities to push lies, conspiracies, and financial scams (11). Here, using the likenesses of trusted and well-known personalities is particularly powerful. Thanks to the past two decades of social media and the pressure to put content online, news anchors and celebrities have unknowingly provided their voices and likenesses for anyone to capture and exploit, now and forever.

Without the use of deepfakes, disinformation has already sowed uncertainty and polarization around everything from our COVID response to our climate change response, our understanding of geopolitical conflicts, and the integrity of elections around the world (12). The introduction of deepfakes of news anchors to this already problematic landscape is sure to only exacerbate our ability to reason about a complex and fast-moving world.

The doctrine of "fair use"—codified into the US Copyright Act of 1976—originated in common law during the 18th and 19th centuries and was designed to both protect creators and not be overly rigid so as to stifle innovation and creativity. In the same way, creators should have some say in how or whether their work is used to train AI systems. This can be accomplished, for example, by inserting into content an imperceptible and resilient digital watermark (13) that can specify everything from copyright ownership to the rules governing its use for training AI systems. There will, of course, be bad actors who choose to ignore these rules. But technological audits of their systems could reveal the presence of training data used without permission (14).

Some regulatory guidelines can also mitigate disruptions. The US Senate, for example, has proposed the NO FAKES Act. If passed, this bill would create new federal intellectual property protections against the misappropriation of a person's voice or likeness, prohibiting the unauthorized use of digital replicas without the informed consent of the depicted individuals. And the recent Digital Services Act passed by the European Union seeks to hold social media platforms accountable for promoting dangerous disinformation.

There is good evidence that the public is uneasy with the role of generative AI in the creative sector. A recent Yahoo Entertainment/YouGov poll (15) found that 61% of US respondents believe that more AI-generated actors is a bad idea, with 29% unsure; and 63% believe that more AI-written scripts is a bad idea, with 28% unsure. Similarly, the US Copyright Office is hearing from creators asking for protection from generative AI (16).

Real Risks

We have focused our discussion on news anchors because they continue to hold a unique position of familiarity and trust in our society. Many other creators, however, are likely to see disruption to their sectors, including writers, editors, photographers, graphic artists, actors, voice-actors, musicians, and the enormous ecosystem that supports these creators. We believe there is a path forward that allows creators to benefit from their past labor, protect their future livelihoods, and leverage this latest exciting technology. In addition to the impact on existing performers, we may also soon see the resurrection of historical figures from Walter Cronkite to Hollywood greats like Sidney Poitier and Katharine Hepburn, raising thorny issues of right to publicity (17).

Importantly, this technology will not reside only with well-resourced networks and Hollywood studios, but will soon become fully democratized. As such, the disruption to the entertainment industry, not to mention the news business, may not stop with writers and performers. The very studios advocating for the use of AI may also find themselves replaced by small teams that will use AI to write and produce content, which can then be distributed and monetized on social media.

As we have shown, these risks are not hypothetical. Generative AI technology is here, becoming increasingly more powerful, and seeing widespread adoption. The time to discuss implications and interventions regarding disruption to the workforce to large-scale disinformation campaigns is now.

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